

PHISHING URL DETECTION REPORT

**MACHINE LEARNING & STREAMLIT WEB APP
PROJECT**

PROJECT REPORT

BY

AKSHAY M

DATA SCIENTIST

**UNDER THE GUIDANCE & SUPERVISION OF
Ms. ARSHA K BABU**

Palakkad, Kerala, 678702

TECHOLAS TECHNOLOGIES

CONTENTS

1. INTRODUCTION

2. ABSTRACT

3. OBJECTIVES

4. METHODOLOGY

5. DATASET OVERVIEW

6. DATA ANALYSIS & EDA VISUALS

7. MODEL DEVELOPMENT & RESULTS

8. STREAMLIT WEB APP DEPLOYMENT

9. CONCLUSION & FUTURE SCOPE

1. INTRODUCTION

The rapid global adoption of digital services is fundamentally transforming online behavior and placing unprecedented pressure on cybersecurity infrastructure. As millions of users interact with web platforms simultaneously, accurately detecting phishing URLs becomes a critical challenge for organizations, security teams, and end-users.

Phishing attacks are highly dynamic—they fluctuate based on obfuscation techniques, domain manipulation, and deceptive keywords. Without accurate threat detection, users risk data breaches, financial loss, and compromised credentials.

Why Phishing Detection Matters

- **For Organizations:** Enables proactive threat mitigation and risk management.
- **For Cybersecurity Teams:** Reduces incident response time and optimizes resource allocation.
- **For End-Users:** Supports a safer browsing experience and provides essential data protection.

The objective of this project is to leverage Deep Learning algorithms to build an accurate phishing URL detection model. This model utilizes an Artificial Neural Network (ANN) to classify malicious links based on extracted lexical and structural properties.

2. ABSTRACT

This project focuses on predicting whether a given URL is legitimate or phishing using an Artificial Neural Network (ANN) built with Keras. The dataset consists of 235,795 URLs evaluated across 21 structural and lexical features, including domain length, obfuscation ratio, HTTPS usage, and keyword presence.

After performing thorough data cleaning, preprocessing, and extensive Exploratory Data Analysis (EDA), a Sequential ANN model was trained and evaluated. The architecture consists of multiple dense layers with dropout applied for regularization. The performance of the model was assessed using standard classification metrics.

The results demonstrate that the neural network achieves an exceptional accuracy of 99.3% on both training and validation splits. Furthermore, a custom extraction logic was designed to perfectly replicate the deployed Streamlit app, ensuring the model evaluates the exact same signals during training and inference.

3. OBJECTIVES

- Analyse URL structural data to identify malicious patterns across different domains and lexical features.
- Perform Exploratory Data Analysis (EDA) to understand class balance, missing values, and HTTPS usage trends.
- Build and evaluate a Keras Artificial Neural Network for accurate phishing detection.
- Replicate inference-time feature extraction to ensure exact consistency with the production app.
- Identify key features contributing to phishing risks using dataset correlation analysis.
- Deploy the insights and model via a Streamlit web application.

4. METHODOLOGY

Step 1: Data Collection

- Dataset used: PhiUSIIL Phishing URL Dataset.
- Total Records: 235,795 rows containing raw URLs and labels.

Step 2: Feature Extraction

- Applied custom Python logic to parse raw URLs into 21 distinct numeric features.
- Extracted features include DomainLength, ObfuscationRatio, IsHTTPS, and specific keyword flags.

Step 3: Data Preprocessing

- Split the dataset into 80/20 training and testing subsets with class stratification.
- Applied standard scaling (StandardScaler) to normalize the extracted numeric features.

Step 4: Exploratory Data Analysis (EDA)

- Analysed class distribution and identified top top-level domains (TLDs).
- Examined HTTPS usage differences between legitimate and phishing URLs.

Step 5: Model Building & Training

- Built a Sequential ANN utilizing 128, 64, 32, and 1 neuron layer structures.
- Included Dropout layers (0.3) for effective regularization.

Step 6: Model Evaluation

- Compiled the model using the Adam optimizer and binary crossentropy loss.
- Evaluated performance over 20 training epochs to ensure smooth convergence.

5. DATASET OVERVIEW

- **Source:** PhiUSIIL Phishing URL Dataset
- **Total Records:** 235,795 rows
- **Features Extracted:** 21 structural and lexical indicators
- **Target Variable:** label (1 for Legitimate, 0 for Phishing)
- **Missing Values:** 0 | **Duplicate Records:** Assessed and handled

Class Distribution:

- Legitimate (Label 1): 57.19% (134,850 records)
- Phishing (Label 0): 42.81% (100,945 records)

6. DATA ANALYSIS & EDA VISUALS

Key Observations

- **HTTPS Usage:** 100% of legitimate sites in the dataset utilize HTTPS, while only 49.22% of phishing sites do, revealing a strong discrepancy.
- **Top TLDs:** The most heavily represented Top-Level Domains are .com (112,554), .org (18,793), and .net (7,097).

- **Feature Correlations:** The target label exhibits a positive correlation with URLSimilarityIndex (0.86) and IsHTTPS (0.60), and a negative correlation with DomainLength (-0.28).
- **URL Length:** Phishing URLs tend to have longer overall string lengths compared to their legitimate counterparts.

7. MODEL DEVELOPMENT & RESULTS

Artificial Neural Network (ANN)

- **Architecture:** Composed of Dense layers (128 -> 64 -> 32 -> 1) with ReLU activation functions and a final Sigmoid output.
- **Regularization:** Implemented 0.3 Dropout layers to prevent overfitting on the training data.
- **Performance:** Reached approximately 99.3% accuracy on both training and validation splits.
- **Loss Metric:** Achieved a smooth minimization of binary crossentropy loss across 20 epochs.

8. STREAMLIT WEB APP DEPLOYMENT

The custom feature extraction function (streamlit_extract) was explicitly designed to mirror the deployed Streamlit application. This structural alignment ensures that the model is trained on the exact same feature distribution it will encounter when processing live URLs in a real-world web deployment. The web app efficiently parses user inputs and runs them through the trained Keras model, providing immediate threat classification.

9. CONCLUSION & FUTURE SCOPE

Conclusion

This project successfully demonstrates the application of Deep Learning techniques for detecting phishing URLs. The integration of custom feature extraction and an Artificial Neural Network provides a highly reliable classification framework. The Sequential

ANN model emerged as a robust classifier, achieving a 99.3% accuracy rate, successfully capturing the non-linear relationships in URL structures.

Future Scope

- Incorporate real-time threat intelligence feeds to improve real-world adversarial robustness.
- Build an ensemble model combining the ANN with gradient boosting techniques like XGBoost.
- Expand the dataset to include a wider variety of recent phishing campaigns and obfuscation strategies.
- Implement continuous monitoring and automated retraining to adapt to new phishing patterns.